Inequalities and equality cases

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Exercise 1. For $x > 0$, prove $x + \frac{1}{x} \geq 2$ and determine equality.

Exercise 2. Find the minimum of $x^2 - 6x + 13$ on $\mathbb{R}$.

Exercise 3. A rectangle has perimeter 24 m. What is its greatest area?

Exercise 4. For real a,b, prove $a^2 + b^2 \geq 2ab$. Must a and b be positive?